Problem 1 (20 points)

A uniform electric field with magnitude E is directed in positive y direction such that $\vec{E} = E\hat{j}$. Consider three points in space with coordinates:
$\vec{r}_a = (0, 0, 0)$, $\vec{r}_b = (0, h, 0)$, and $\vec{r}_c = (d, 0, 0)$

(a) (10pts.) Find the potential difference $V_a - V_b$. Which point is at a higher potential?

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E\, dy = Eh \Rightarrow \text{positive}$$

"a" is at a higher potential.

(b) (5pts.) Find the work that must be done by an external agent to move a positive charge q from b to a, with constant velocity.

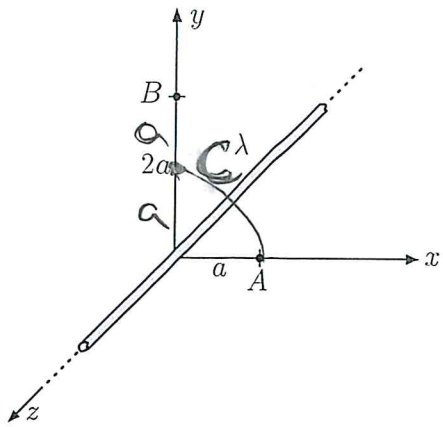
$$W_{ext}(b \to a) = \Delta U = U_a - U_b = q(V_a - V_b)$$
$$= qEh \Rightarrow \text{positive}$$

(c) (5pts.) Find the work that must be done by an external agent to move a positive charge q from a to c, with constant velocity.

$$V_a - V_c = 0$$
$$W_{a \to c} = W_{c \to a} = 0$$

Problem 2 (20 points)

An infinitely long positive line charge having a uniform charge per length λ is located along the $z-$axis as shown in the figure.



(a)(10 pts.) Using Gauss' Law, find the electric field vector at point A.

Consider a cylinder of length L, radius r

$\int \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$ $\quad q_{in} = \lambda L \Rightarrow E\, 2\pi r L = \frac{\lambda L}{\varepsilon_0}$

$\Rightarrow E = \frac{\lambda}{2\pi r \varepsilon_0}$ at $r = a$ $\quad \boxed{\vec{E}_A = \frac{\lambda}{2\pi a \varepsilon_0} \uparrow}$

(b)(10 pts.) Find the work done by the electric field to move a positive charge from poin A to point B.

$W = -\Delta U = -q(V_{fin} - V_{in})$

Work is independent of path. Let us follow the path $A \to C \to B$.

$W_{A\to C} = 0$, since $V_A - V_C = 0$

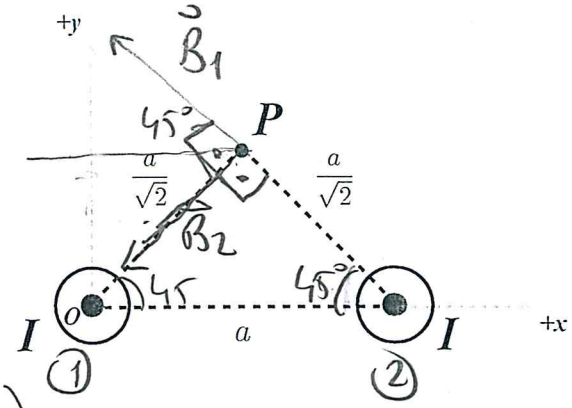
$W_{C\to B} = -q(V_B - V_C)$

$V_B - V_C = -\int_C^B \vec{E} \cdot d\vec{s} = -\int_a^{2a} \frac{\lambda}{2\pi r \varepsilon_0} dr = -\frac{\lambda}{2\pi\varepsilon_0} \ln(2)$

$W_{C\to B} = \frac{q\lambda}{2\pi\varepsilon_0} \ln(2) \Rightarrow$ work done by the electric field.

Problem 3 (20 points)

Each of the two long parallel wires shown in the figure carry current I directed out of the page. Find the magnetic field vector at point P. Express your results in terms of unit vectors.



$$B = \frac{\mu_0 I}{2\pi r} \quad \text{(for a long wire)}$$

$$B_1 = \frac{\mu_0 I}{2\pi\left(\frac{a}{\sqrt{2}}\right)} = \frac{\mu_0 I}{\sqrt{2}\pi a} \Rightarrow \text{magnitude of the magnetic field vector.}$$

$$\vec{B}_1 = B_1(-\cos 45\,\hat{i} + \sin 45\,\hat{j})$$

$$\vec{B}_1 = \frac{\mu_0 I}{\sqrt{2}\pi a}\left(-\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right) = \frac{\mu_0 I}{2\pi a}(-\hat{i} + \hat{j})$$

$$B_2 = \frac{\mu_0 I}{2\pi\left(\frac{a}{\sqrt{2}}\right)} = \frac{\mu_0 I}{\sqrt{2}\pi a} \Rightarrow \text{magnitude of the magnetic field vector}$$

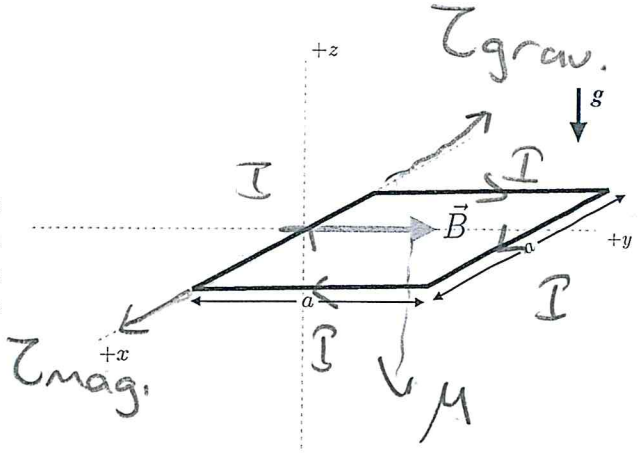
$$\vec{B}_2 = B_2(-\cos 45\,\hat{i} - \sin 45\,\hat{j})$$

$$\vec{B}_2 = \frac{\mu_0 I}{2\pi a}(-\hat{i} - \hat{j})$$

$$\vec{B}(P) = \vec{B}_1 + \vec{B}_2 = -\frac{\mu_0 I}{\pi a}\hat{i}$$

Problem 4 (20 points)

A square wire loop with side length a, carrying current I, has mass M. The loop is hinged such that it can rotate around the x-axis freely. A uniform magnetic field is applied in the $+y$ direction and its magnitude is B. If the loop is in the equilibrium,



(a)(5 pts.) show the direction of the current.

Torque (gravity) $= Mg\left(\frac{a}{2}\right)(-\hat{i})$ (rotates the loop clockwise around the "x" axis)

Torque (magnetic) should be in $(+\hat{i})$ direction

Torque (magnetic) $= \vec{\mu} \times \vec{B} = \vec{\mu} \times (B\hat{j})$

$\Rightarrow \vec{\mu}$ is in $(-\hat{k})$ direction)

$\Rightarrow$ The current travels clock-wise around the loop.

(b)(15 pts.) Find current I in terms of M, B, a and g (gravitational acceleration).

In equilibrium $\tau_{net} = 0$

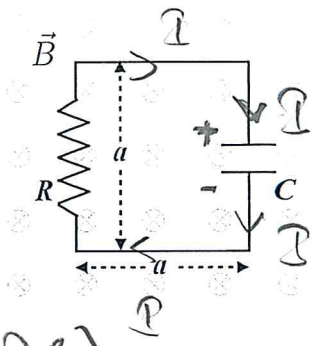
$\tau_{grav} = Mg\left(\frac{a}{2}\right)(-\hat{i})$

$\tau_{mag} = \vec{\mu} \times \vec{B} = I \cdot a^2 B\,(+\hat{i})$

$\tau_{net} = 0 \Rightarrow I a^2 B = Mg\frac{a}{2} \Rightarrow \boxed{I = \frac{Mg}{2aB}}$

Problem 5 (20 points)

A uniform magnetic field changes at a constant rate $dB/dt = -K$, where K is a positive constant. A square loop of wire of side length a containing a resistance R and a capacitance C is placed with its plane normal to the field.



(a)(5 pts.) Find the induced emf in the loop.

$$\mathcal{E} = -\frac{\partial \Phi_B}{\partial t} = -\frac{\partial}{\partial t}(a^2 B) = a^2\left(-\frac{\partial B}{\partial t}\right)$$

$$= a^2 K$$

(b)(10 pts.) Find the charge Q on the capacitor when it is fully charged.

$Q = C\mathcal{E}$ when the capacitor is fully charged.

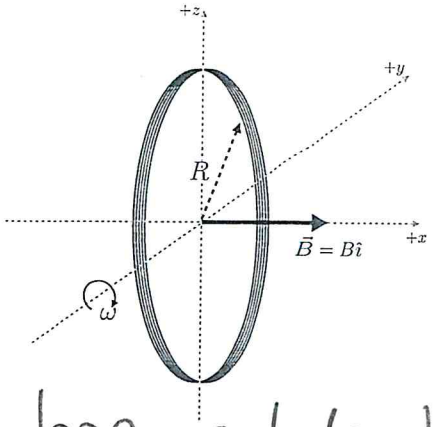
$Q_{max} = Ca^2K$

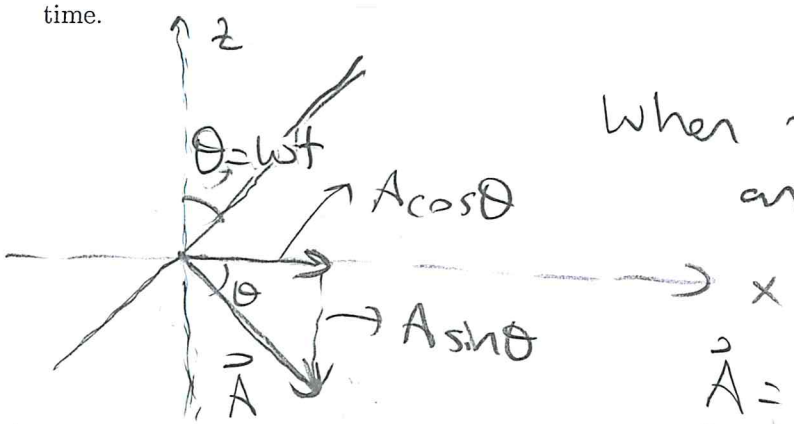
(c)(5 pts.) Which plate, upper or lower, is at the higher potential when the capacitor is fully charged? Please explain why.

The magnetic field is into the page and the magnetic flux decreases. The induced current travels counter clockwise around the loop. The potential across the capacitor will decrease in the direction of the current. Therefore the upper plate is at the higher potential.

Problem 6 (20 points)

A circular wire loop of N turns, each with radius R, is rotated with a constant angular velocity ω around the y-axis in the presence of a uniform magnetic field $\vec{B} = B\hat{\imath}$ as shown in the figure. If the loop lies in the $y - z$ plane at $t = 0$, find the induced e.m.f as a function of time.





When the loop rotates by an angle $\theta = \omega t$

$$\vec{A} = A(\cos\theta\,\hat{\imath} - \sin\theta\,\hat{k})$$

$$\vec{B} = B\hat{\imath}$$

$$\phi_B = N(\vec{A}\cdot\vec{B}) = NBA\cos(\omega t)$$

$$\mathcal{E} = -\frac{\partial\phi_B}{\partial t} = \underline{\underline{NBA\omega\sin(\omega t)}}$$